

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1420

Port Saint Joe 138 kV
April 20, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1420. The Interconnection Customer proposes a 87.2 MW solar photovoltaic facility interconnecting at the Port Saint Joe 138 kV substation in the Virginia area. The request was received on March 07, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 513 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,242,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Rocky Reach / Idaho Springs 138 kV Line	Sideburn / Yavapai Substation 138 kV line	117.3	255	15.4
Burke-Sideburn / Rocky Reach 138 kV	Yavaai Substation / Sideburn 138 kV Line	115.0	238	15.4
Glen Carlyn Substation / Centennial 138 kV	Idylwood Substation / Gallows Rd 138 kV	115.0	278	7.4
Yavapai / Sideburn 138 kV	Colonial / Idaho Springs 138 kV	113.1	292	23.5
Occoquan / Covanta Fairfax Energy 138 kV Line	Fairfax / Jeune Landing 138 kV	111.7	212	10.1
Centennial / Annandale Substation 138 kV Line	Idylwood / Gallows Rd 138 kV line	110.9	345	7.4
Gallows Rd / Idylwood 138 kV	Centennial / Annadale Substation 138 kV	110.3	323	8.7
Yavapai Substation / Burke-Sideburn 138 kV	Sideburn / Yavapai 138 kV line	106.2	223	15.4
Falls Church / Decatur Energy Ctr 138 kV	Centennial / Annandale 138 kV line	104.9	204	8.7
South Houston Substation / Peavy Fall Hy 138 kV L	Fillmore / S Houston 138 kV L	103.4	261	19.2
Luna / Peavy Fall Hy 138 kV	S Houston / Fillmore 138 kV	101.9	313	19.2
Yavapai / Luna Substation 138 kV	Fillmore / S Houston 138 kV L	101.1	353	19.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rocky Reach / Idaho Springs 138 kV Line	Rebuild line with higher-capacity conductor	\$1,333,000	21% (\$285,000)
Burke-Sideburn / Rocky Reach 138 kV	Rebuild line with higher-capacity conductor	\$1,348,000	21% (\$288,000)
Glen Carlyn Substation / Centennial 138 kV	Rebuild line with higher-capacity conductor	\$1,588,000	9% (\$142,000)
Yavapai / Sideburn 138 kV	Rebuild line with higher-capacity conductor	\$1,564,000	27% (\$420,000)
Occoquan / Covanta Fairfax Energy 138 kV	Rebuild line with higher-capacity conductor	\$3,784,000	9% (\$344,000)
Centennial / Annandale 138 kV	Rebuild line with higher-capacity conductor	\$1,427,000	9% (\$128,000)
Gallows Rd / Idylwood 138 kV	Rebuild line with higher-capacity conductor	\$774,000	9% (\$71,000)
Yavapai / Burke-Sideburn 138 kV	Reconductor line	\$1,141,000	21% (\$244,000)
Falls Church / Decatur Energy Ctr 138 kV Line	Reconductor line	\$399,000	9% (\$36,000)
Peavy Fall Hy / S Houston 138 kV	Reconductor line	\$432,000	20% (\$87,000)
Luna / Peavy Fall Hy 138 kV Line	Reconductor line	\$561,000	20% (\$113,000)
Yavapai / Luna Substation 138 kV	Reconductor line	\$414,000	20% (\$84,000)
	Total allocated to project		\$2,242,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.